ARJUNA JEE 2027

Lecture -06 | Mathematics

BINOMIAL THEOREM





By- Sachin Jakhar Sir

Topics *to be covered*

A **Binomial Series**

B **Question Practice**

C

D



H.W. Discussion

Question [JEE Mains-2020]

If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+\ldots+x^{2n})(1-x+x^2-x^3+\ldots+x^{2n})$ is 61, then n is equal to _____.

$(1+x+x^2+\ldots+x^{2n})(1-x+x^2-x^3+\ldots+x^{2n}) = a_0+a_1x^1+a_2x^2+\ldots+a_{4n}x^{4n}$

47%

$x=1$ $\quad (2n+1)\times(1) = a_0+a_1+a_2+a_3+a_4+\ldots\ldots$

$x=-1$ $\quad 1\times(2n+1) = a_0-a_1+a_2-a_3+a_4-a_5\ldots\ldots$

add

$61 = 2(2n+1) = 2(a_0+a_2+a_4+\ldots\ldots)$

$30 = \frac{60}{2} = n$

RAPID FIRE DIBY'S

DIBY-09

The coefficient of x^{49} in the expansion of $(x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right)\ldots\left(x-\frac{1}{2^{49}}\right)$ is equal to –

50 Brackets.

(A) $-2\left(1-\frac{1}{2^{50}}\right)$

(B) +ve coefficient of x

(C) -ve coefficient of x

(D) $-2\left(1-\frac{1}{2^{49}}\right)$

$x^{50} \ominus \left(1+\frac{1}{2}+\frac{1}{2^2}+\ldots+\frac{1}{2^{49}}\right)x^{49} + \left(1\cdot\frac{1}{2}+1\cdot\frac{1}{2^2}+\ldots\right)x^{48}$

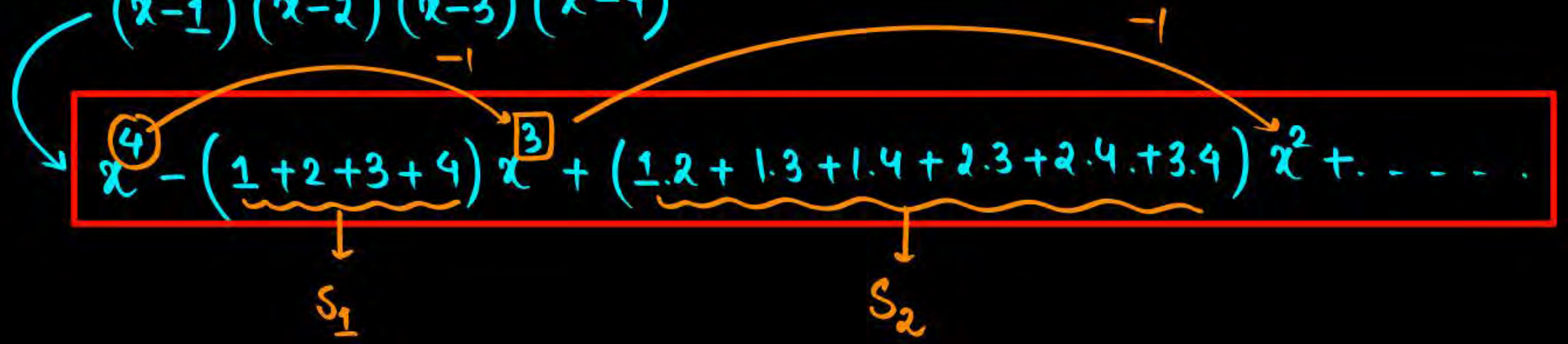
G.P. $a=1$, $r=\frac{1}{2}$, $n=50$

[Ans. : (A)]

Chota Roop :

4-Brackets

$(x-1)(x-2)(x-3)(x-4)$



RAPID FIRE DIBY'S

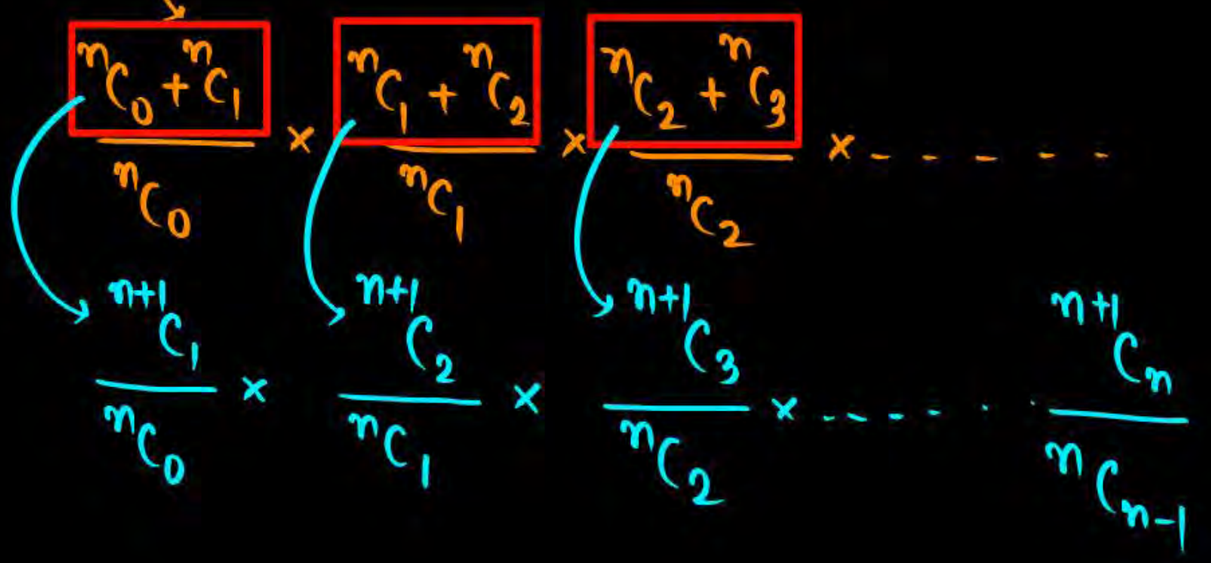
DIBY-03

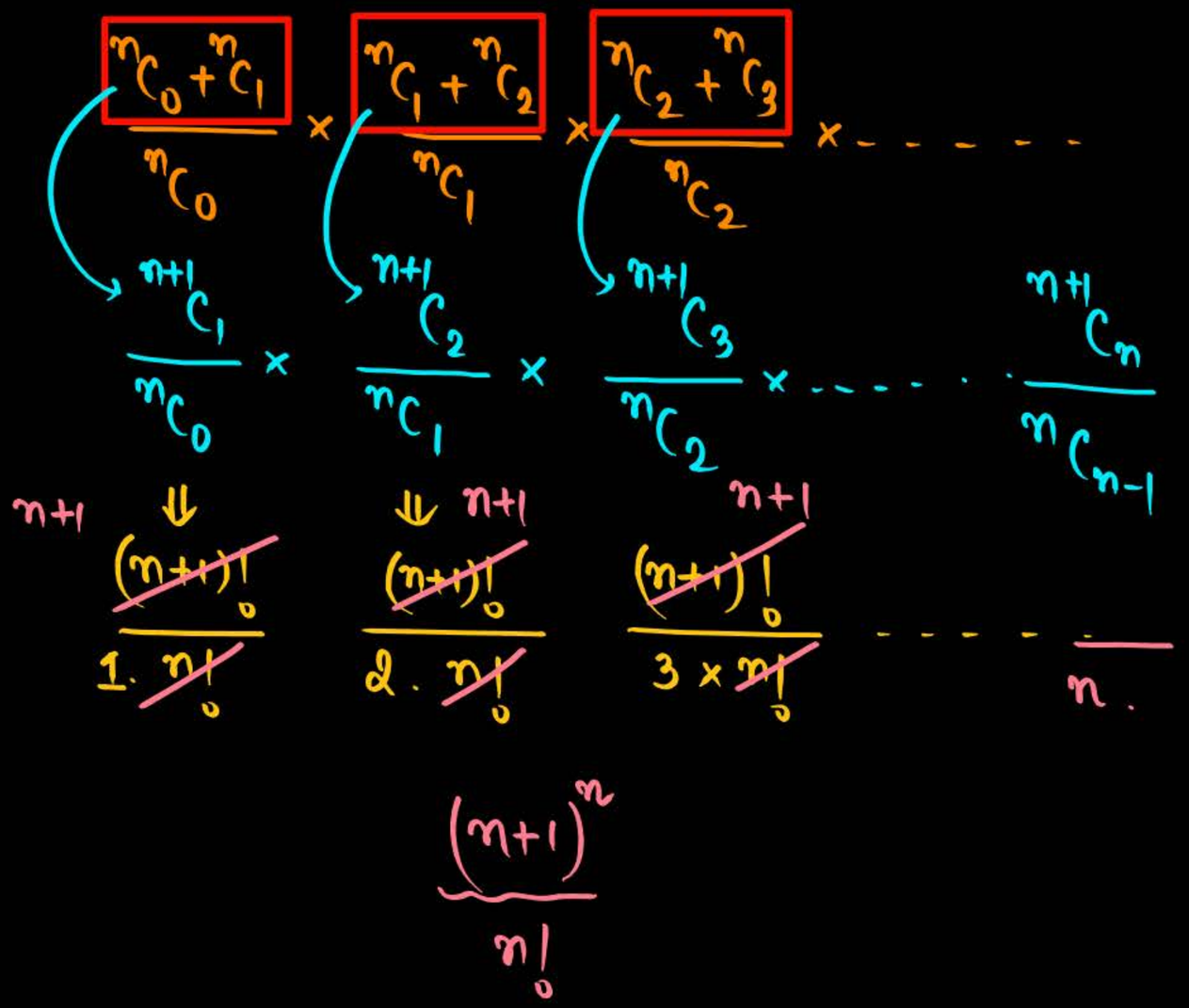
Evaluate the value of expression : $\left(\frac{C_0+C_1}{C_0}\right)\left(\frac{C_1+C_2}{C_1}\right)\left(\frac{C_2+C_3}{C_2}\right)\ldots\left(\frac{C_{n-1}+C_n}{C_{n-1}}\right)$

(where $C_r = \binom{n}{r} = {}^nC_r$**)**

(A) $\frac{(n+1)^n}{n!}$ **(B)** $\frac{n^{n+1}}{(n+1)!}$ **(C)** $\frac{(n+1)^{n-1}}{n!}$ **(D)** $\frac{(n+1)^n}{(n-1)!}$

[Ans. : (A)]




$^nC_0 + {}^nC_1$
$^nC_1 + {}^nC_2$
$^nC_2 + {}^nC_3$
$^nC_0$
$^nC_1$
$^nC_2$
$\frac{^{n+1}C_1}{^nC_0} \times \frac{^{n+1}C_2}{^nC_1} \times \frac{^{n+1}C_3}{^nC_2} \times \cdots \frac{^{n+1}C_n}{^nC_{n-1}}$
$n+1$
$\frac{(n+1)!}{1.\,n!} \quad \frac{(n+1)!}{2.\,n!} \quad \frac{(n+1)!}{3 \times n!} \quad \cdots \quad \frac{}{n.}$
$\frac{(n+1)^n}{n!}$

Question [AIEEE-2007]

79%

The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots + {}^{20}C_{10}$ is

A $\frac{1}{2}{}^{20}C_{10}$

B 0

C $-{}^{20}C_{10}$

D ${}^{20}C_{10}$

$${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots - {}^{20}C_9 + {}^{20}C_{10} - {}^{20}C_{11} - \ldots + {}^{20}C_{18} - {}^{20}C_{19} + {}^{20}C_{20} = 0$$

$$- {}^{20}C_9 \quad + {}^{20}C_2 \quad - {}^{20}C_1 + {}^{20}C_0$$

$$\left\{{}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - \ldots - {}^{20}C_9 + {}^{20}C_{10}\right\} = \frac{{}^{20}C_{10}}{2}$$

Question [JEE Mains-2023]

Suppose $\sum_{r=0}^{2023} r^2 \cdot {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is

80%

$r \cdot r \cdot {}^{n}C_r = r \cdot n \cdot {}^{n-1}C_{r-1}$

$= n\left((r-1)+1\right){}^{n-1}C_{r-1}$ ★★

$= n\left((r-1)\cdot {}^{n-1}C_{r-1} + {}^{n-1}C_{r-1}\right)$

$n(n-1)\sum_{r=2}^{n} {}^{n-2}C_{r-2} + n\sum_{r=1}^{n} {}^{n-1}C_{r-1} = n(n-1)\,2^{n-2} + n\left(2^{n-1}\right)$

$2023 \times 2^{2022}\left\{1011 + 1\right\}$

$2023 \times 2022 \times 2^{2021} + 2023 \times 2^{2022}$

(2×1011)

Question [JEE Mains-2021]

Let ${}^nC_r$ denote the binomial coefficient of x^r in the expansion of $(1 + x)^n$.

If $\sum_{k=0}^{10} (2^k + 3k)\,{}^{10}C_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$, $\alpha, \beta \in R$, $\alpha + \beta$ is equal to _____.



Type-02: *When Binomial Coefficients are divided by a constant*

Question

Show that : $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

$$\frac{{}^{n}C_{r}}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$$

Question [JEE Mains-2024]

If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \cdots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $gcd(n, m) = 1$, then $n + m$ is equal to

2041

76%

$$\sum_{r=1}^{9} \frac{{}^{11}C_r}{r+1} = \sum_{r=1}^{9} \frac{{}^{12}C_{r+1}}{12} = \frac{1}{12}\left(\sum_{r=1}^{9} {}^{12}C_{r+1}\right)$$

$$= \frac{1}{12}\left({}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 + {}^{12}C_4 + \cdots + {}^{12}C_{10} + {}^{12}C_{11} + {}^{12}C_{12} - \left({}^{12}C_0 - {}^{12}C_1 - {}^{12}C_{11} - {}^{12}C_{12}\right)\right)$$

$$= \frac{1}{12}\left(2^{12} - 26\right) = \frac{4070}{12} = \frac{2035}{6}$$

Question [JEE Mains-2024]

M-2 (Integration)

If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \cdots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $gcd(n, m) = 1$, then $n + m$ is equal to

$$\int_0^x (1+x)^{11}\,dx = \int_0^x {}^{11}C_0\,dx + \int_0^x {}^{11}C_1\,x^1\,dx + \int_0^x {}^{11}C_2\,x^2\,dx + \int_0^x {}^{11}C_3\,x^3\,dx + \cdots\cdots$$

$$\left.\frac{(1+x)^{12}}{12}\right|_0^x = {}^{11}C_0\left(x\right)_0^x + {}^{11}C_1\left(\frac{x^2}{2}\right)_0^x + {}^{11}C_2\left(\frac{x^3}{3}\right)_0^x + \cdots\cdots$$

$$\frac{(1+x)^{12}}{12} - \frac{1}{12} = {}^{11}C_0\,x + \frac{{}^{11}C_1}{2}x^2 + \frac{{}^{11}C_2}{3}x^3 + \frac{{}^{11}C_3}{4}x^4 + \cdots\cdots$$

$x = 1$

$$\frac{2^{12}}{12} - \frac{1}{12} = {}^{11}C_0 + \boxed{\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \frac{{}^{11}C_3}{4} + \cdots\cdots + \frac{{}^{11}C_9}{10}} + \frac{{}^{11}C_{10}}{11} + \frac{{}^{11}C_{11}}{12}$$

$$\frac{2^{12}-1}{12} - \frac{1}{12} - 1 - 1$$

$$\frac{2^{12}-2-24}{12}$$

$$\frac{2^{12}-26}{12}$$

Question [JEE Mains-2023]

If $\frac{1}{n+1}\,{}^{n}C_{n}+\frac{1}{n}\,{}^{n}C_{n-1}+\ldots+\frac{1}{2}\,{}^{n}C_{1}+\,{}^{n}C_{0}=\frac{1023}{10}$, then n is equal to

Ans: A

A 9

B 6

C 7

D 8

QIBY - A

Question

${}^{n}C_r = C_r$.



Show that : $\frac{{}^{n}C_0}{2} + \frac{{}^{n}C_1}{3} + \frac{{}^{n}C_2}{4} + \ldots + \frac{{}^{n}C_n}{n+2} = \frac{n \cdot 2^{n+1}+1}{(n+1)(n+2)}$.

$$\# \text{ LHS} = \sum_{r=0}^{n} \frac{{}^{n}C_r}{(r+2)(r+1)} = \sum_{r=0}^{n} \frac{{}^{n+1}C_{r+1}}{(n+1)(r+2)} \cdot \frac{(r+1)}{(r+1)}$$

$$= \sum_{r=0}^{n} \frac{(r+1)\,{}^{n+2}C_{r+2}}{(n+1)(n+2)}$$

$$= \frac{1}{(n+1)(n+2)} \sum_{r=0}^{n} (r+1)\,{}^{n+2}C_{r+2}$$

$$\#\ RHS = \frac{1}{(n+1)(n+2)} \sum_{r=0}^{n} (r+2-1)\ {}^{n+2}C_{r+2}$$

$$\#\ RHS = \frac{1}{(n+1)(n+2)} \sum_{r=0}^{n} \left((r+2)\ {}^{n+2}C_{r+2} - {}^{n+2}C_{r+2} \right)$$

$$\#\ RHS = \frac{1}{(n+1)(n+2)} \sum_{r=0}^{n} \left((n+2)\ {}^{n+1}C_{r+1} - {}^{n+2}C_{r+2} \right)$$

$$\#\ RHS = \frac{1}{(n+1)(n+2)} \left\{ (n+2) \sum_{r=0}^{n} {}^{n+1}C_{r+1} - \sum_{r=0}^{n} {}^{n+2}C_{r+2} \right\}$$

$$(n+2)\left(2^{n+1} - {}^{n+1}C_0\right)$$

$${}^{n+2}C_2 + {}^{n+2}C_3 + \ldots + {}^{n+2}C_{n+2}$$

$$\Downarrow$$

$$\left(2^{n+2} - {}^{n+2}C_0 - {}^{n+2}C_1\right)$$





100% Ques^{n}. आएगा

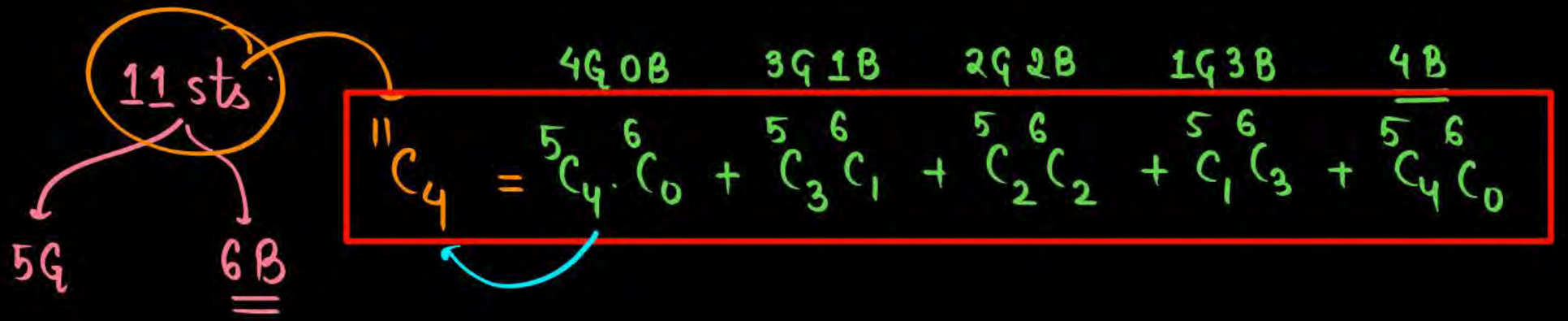
Type-03 : *When Binomial Coefficients in Multiplication (Bino-binomial Series)*

Question Forming : Two different expansion are multiplied and coefficient of some power of x is asked.

Example

P4C

Let in a class of 11 students there are five girls and six boys, Then find number of ways of selecting any four students is_____.



FORMATION OF BINO-BINOMIAL SERIES

$$(1+x)^5 \cdot (1+x)^6 = ({}^5C_0 + {}^5C_1x^1 + {}^5C_2x^2 + \cdots + {}^5C_5x^5)$$
$$\cdot ({}^6C_0 + {}^6C_1x^1 + {}^6C_2x^2 + \cdots + {}^6C_6x^6)$$
$$= {}^5C_0^6C_0 + ({}^5C_1{}^6C_0 + {}^5C_0^6C_1)x + ({}^5C_2^6C_0 + {}^5C_1^6C_1 + {}^5C_0^6C_2)x^2 + \cdots$$
$$\ldots + ({}^5C_0^6C_6 + {}^5C_1^6C_5 + {}^5C_2^6C_4 + {}^5C_3^6C_3 + {}^5C_4^6C_2 + {}^5C_5^6C_1)x^6 + \cdots$$

??

$${}^{11}C_6.$$

Coeff. of x^6 in $(1+x)^5(1+x)^6 = {}^{11}C_6$

Coeff of x^6 in $(1+x)^{11}$

$\hookrightarrow {}^{11}C_r x^r$

DIRECT FORMULA FOR BINO-BINOMIAL SERIES

1. upper suffix is same in each term. ✓
2. Sum of lower suffix in each term is equal ✓
3. "all" terms are presents in add". ✓

Direct Ans. $\text{sum of upper suffix}\ C\ \text{sum of lower suffix}$

Blue बादल (KKPNCC)
अगर ये conditions fulfill ना हो तो
Direct formula नहीं लगेगा।
THEN "GO BY THE PROCESS"

$$S = \binom{7}{0}^2 + \binom{7}{1}^2 + \binom{7}{2}^2 + \cdots\cdots + \binom{7}{7}^2$$

$$S = {}^7C_0\,{}^7C_0 + {}^7C_1\,{}^7C_1 + {}^7C_2\,{}^7C_2 + \cdots\cdots + {}^7C_7\,{}^7C_7$$

$$S = {}^7C_0\,{}^7C_7 + {}^7C_1\,{}^7C_6 + {}^7C_2\,{}^7C_5 + \cdots\cdots + {}^7C_7\,{}^7C_0 = {}^{14}C_7$$

Example

$$S = ({}^{n}C_0)^2 + ({}^{n}C_1)^2 + ({}^{n}C_2)^2 + ... + ({}^{n}C_n)^2$$

$$\rightarrow {}^{n}C_0\,{}^{n}C_n + {}^{n}C_1\,{}^{n}C_{n-1} + {}^{n}C_2\,{}^{n}C_{n-2} + \cdots + {}^{n}C_n\,{}^{n}C_0 = {}^{2n}C_n \checkmark$$

Question [JEE Mains-2023]

The value of $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ is

(

$${}^{22}C_0 \, {}^{23}C_{23} + {}^{22}C_1 \, {}^{23}C_{22} + {}^{22}C_2 \, {}^{23}C_{21} + \cdots\cdots + {}^{22}C_{22} \, {}^{23}C_1 = {}^{45}C_{23}.$$

A $^{44}C_{23}$

B $^{45}C_{23}$

C $^{44}C_{22}$

D $^{45}C_{24}$

Question [JEE Mains-2024]

$$\frac{{}^{n}C_{k+1}}{k+2} = \frac{{}^{n+1}C_{k+2}}{n+1}$$

Let $\alpha = \sum_{k=0}^{n} \left(\frac{({}^{n}C_{k})^2}{k+1} \right)$ and $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^{n}C_{k}\,{}^{n}C_{k+1}}{k+2} \right)$. If $5\alpha = 6\beta$, then n equals

$$\alpha = \sum_{k=0}^{n} \frac{{}^{n}C_{k} \cdot {}^{n}C_{k}}{(k+1)}$$

$$\alpha = \frac{1}{(n+1)} \sum_{k=0}^{n} {}^{n}C_{k}\;{}^{n+1}C_{k+1}$$

$$= {}^{n}C_{n}\;{}^{n+1}C_{1} + {}^{n}C_{n-1}\;{}^{n+1}C_{2} + {}^{n}C_{n-2}\;{}^{n+1}C_{3} + \ldots + {}^{n}C_{0}\;{}^{n+1}C_{n+1} = \frac{{}^{2n+1}C_{n+1}}{n+1}$$

$$\beta = \frac{1}{n+1} \sum_{k=0}^{n-1} {}^{n}C_{k}\;{}^{n+1}C_{k+2}$$

$$= {}^{n}C_{n}\;{}^{n+1}C_{2} + {}^{n}C_{n-1}\;{}^{n+1}C_{3} + {}^{n}C_{n-2}\;{}^{n+1}C_{4} + \ldots + {}^{n}C_{1}\;{}^{n+1}C_{n+1}$$

$$\beta = \frac{{}^{2n+1}C_{n+2}}{n+1}$$

$$\beta = \frac{{}^{2n+1}C_{n+2}}{n+1}$$

$$\alpha = \frac{{}^{2n+1}C_{n+1}}{n+1}$$

$$5\alpha = 6\beta$$

$$5 \cdot \frac{{}^{2n+1}C_{n+1}}{n+1} = 6 \frac{{}^{2n+1}C_{n+2}}{n+1}$$

$$5 \cdot \frac{(2n+1)!}{(n+1)!\, n!} = 6 \times \frac{(2n+1)!}{(n+2)!\,(n-1)!}$$

$n! = n(n-1)!$

$(n+2)! = (n+2)(n+1)!$

$$\frac{5}{n} = \frac{6}{n+2}$$

$$5n + 10 = 6n$$

$$10 = n$$

Question [JEE Mains-2019]

The value of r for which

$^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \ldots + {}^{20}C_0 \cdot {}^{20}C_r$ is maximum is _____.

A 15

B 10

C 20

D 11

$^{40}C_r \Big|_{\max.} \Rightarrow r = \frac{40}{2} = 20.$

Question [JEE (Adv.)-2018]

Let $X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + \ldots\ldots + 10({}^{10}C_{10})^2$ where $r \in \{1, 2, \ldots, 10\}$ & ${}^{10}C_r$ denote binomial coefficients. Then the value of $\frac{1}{1430}X$ is_____.

$$X = \sum_{r=1}^{10} r \cdot {}^{10}C_r \cdot {}^{10}C_r$$

$$X = 10\left(\sum_{r=1}^{10} {}^{9}C_{r-1}\,{}^{10}C_r\right) = 10\left({}^{9}C_9\,{}^{10}C_1 + {}^{9}C_8\,{}^{10}C_2 + {}^{9}C_7\,{}^{10}C_3 + \ldots\ldots + {}^{9}C_0\,{}^{10}C_{10}\right)$$

$$X = 10\left({}^{19}C_{10}\right)$$

$$\frac{1}{1430} \times 10 \times \frac{19!}{10!\,9!} = \frac{1}{143}\left(\frac{19.18.17.16.15.14.13.12.11}{9.8.7.6.5.4.3.2.1}\right) = 646$$

Question [IIT-JEE-2010]

For $r = 0, 1, \ldots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions $(1 + x)^{10}, (1 + x)^{20}$ and $(1 + x)^{30}$.

Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to



A $B_{10} - C_{10}$

B $A_{10}\left(B_{10}^2 - C_{10}A_{10}\right)$

C 0

D $C_{10} - B_{10}$

Example

Notes

$$S = {}^{10}C_0\,{}^{11}C_1 - {}^{10}C_1\,{}^{11}C_2 + {}^{10}C_2\,{}^{11}C_3 - {}^{10}C_3\,{}^{11}C_4 + \ldots\ldots\ldots\ldots + {}^{10}C_{10}\,{}^{11}C_{11} = {}^{21}C_{11} \times$$

$$\hookrightarrow = -{}^{10}C_5 .$$

$$(1-x)^{10}(1+x)^{11}$$

sum of lower suffix in each term

Coeff. of x^{11} in $(1-x)^{10}(1+x)^{11}$

$$(1-x)^{10}(1+x)^{10}\cdot(1+x)$$

$$(1-x^2)^{10}\cdot(1+x)$$

x^{11} $\quad$ x^0

$${}^{10}C_5(-1)^5 x^{10} \times 1x^1 = \left(-{}^{10}C_5\right)x^{11}$$

$${}^{10}C_r(-1)^r \cdot x^{2r}, \quad r = 5$$

Type-04: *Mixed Profile*

Question [JEE Mains-2019]

If $\sum_{r=0}^{25}\{{}^{50}C_r \cdot {}^{50-r}C_{25-r}\} = K({}^{50}C_{25})$, then K is equal to

A 2^{25}

$\rightarrow {}^{50}C_0\,{}^{50}C_{25} + {}^{50}C_1 \cdot {}^{49}C_{24} + {}^{50}C_2 \cdot {}^{48}C_{23} + \ldots\ldots$ (not a B.B. series for which direct result can be used)

B $2^{25} - 1$

C 2^{24}

D $(25)^2$

Type-04: *Mixed Profile*

Question [JEE Mains-2019]

If $\sum_{r=0}^{25}\{{}^{50}C_r \cdot {}^{50-r}C_{25-r}\} = K({}^{50}C_{25})$, then K is equal to

A 2^{25}

B $2^{25}-1$

C 2^{24}

D $(25)^2$

$$\frac{50!}{r!\,(50-r)!} \times \frac{(50-r)!}{(25-r)!\,25!} = \frac{50!}{25!\,25!} \cdot \frac{25!}{r!\,(25-r)!}$$

$$\sum_{r=0}^{25} {}^{50}C_{25} \cdot {}^{25}C_r = {}^{50}C_{25}\left(\sum_{r=0}^{25} {}^{25}C_r\right) = {}^{50}C_{25} \cdot 2^{25}$$

$${}^{25}C_0 + {}^{25}C_1 + \ldots + {}^{25}C_{25}$$

If $\sum_{r=0}^{1000} \frac{\left({}^{1000}C_r\right)}{\left({}^{2019}C_r\right)} = \frac{m}{n}$, ($m$ and n are relatively prime positive integers)

then $m + n =$ _____.

$$\frac{{}^{2020}C_{1020}}{{}^{2019}C_{1019}}$$

$$\frac{1000!\ r!\ (2019-r)!}{r!\ (1000-r)!\ 2019!\ 1019!} \times 1019! = \frac{1000! \times 1019!}{2019!} \times {}^{2019-r}C_{1019}$$

$$\frac{1}{{}^{2019}C_{1019}}$$

$$\sum_{r=0}^{1000} \frac{{}^{2019-r}C_{1019}}{{}^{2019}C_{1019}} = \frac{1}{{}^{2019}C_{1019}} \sum_{r=0}^{1000} {}^{2019-r}C_{1019}$$

Hockey stick id.

$${}^{2019}C_{1019} + {}^{2018}C_{1019} + {}^{2017}C_{1019} + \ldots + {}^{1019}C_{1019}$$

Acidic Question

$$S = 3 \cdot {}^{10}C_0 + 3^2 \cdot \frac{{}^{10}C_1}{2} + 3^3 \cdot \frac{{}^{10}C_2}{3} + \ldots + 3^{11} \cdot \frac{{}^{10}C_{10}}{11}$$



DO IT BY YOURSELF (DIBY)

NUMBER OF RATIONAL OR IRRATIONAL TERMS

DIBY-28

If n is the number of irrational terms in the expansion of $\left(3^{1/4}+5^{1/8}\right)^{60}$, then $(n-1)$ is divisible by [JEE Mains-2021]

(A) 26 (B) 30 (C) 8 (D) 7

[Ans. : (A)]

DIBY-29

The sum of all rational terms in the expansion of $\left(2^{1/5}+5^{1/3}\right)^{15}$ is equal to [JEE Mains-2024]

(A) 3133 (B) 633 (C) 931 (D) 6131

[Ans. : (A)]

DIBY-30

The Least value of n for which the number of integral terms in the Binomial expansion of $\left(\sqrt[3]{7}+\sqrt[12]{11}\right)^{n}$ is 183, is [JEE Mains-2025]

(A) 2148 (B) 2184 (C) 2196 (D) 2172

[Ans. : (B)]

NUMBER OF RATIONAL OR IRRATIONAL TERMS

DIBY-31

The number of integral terms in the expansion of $\left(5^{1/2}+7^{1/8}\right)^{1016}$ is

[JEE Mains-2025]

(A) 127 (B) 130 (C) 129 (D) 128

[Ans. : (D)]

DIBY-32

The total number of irrational terms in the binomial expansion of $(7^{1/5}-3^{1/10})^{60}$ is

[JEE Mains-2019]

(A) 55 (B) 49 (C) 48 (D) 54

[Ans. : .....]

DIBY-33

The number of rational terms in the binomial expansion of $\left(4^{1/4}+5^{1/6}\right)^{120}$ is

[JEE Mains-2021]

[Ans. : 21]

NUMBER OF RATIONAL OR IRRATIONAL TERMS

DIBY-34

The sum of all those terms which are rational numbers in the expansion of $\left(2^{1/3}+3^{1/4}\right)^{12}$ is [JEE Mains-2021]

(A) 89 (B) 35 (C) 27 (D) 43

[Ans. : (D)]

DIBY-35

Number of integral terms in the expansion of $\left\{7^{(1/2)}+11^{(1/6)}\right\}^{824}$ is equal to [JEE Mains-2024]

[Ans. : 138]

DIBY-36

The sum of all rational terms in the expansion of $\left(2+\sqrt{3}\right)^{8}$ is [JEE Mains-2025]

(A) 16923 (B) 3763 (C) 33845 (D) 18817

[Ans. : (D)]

FINDING COEFFICIENTS

DIBY-37

The coefficient of x^{101} in the expression

$(5+x)^{500}+x(5+x)^{499}+x^2(5+x)^{498}+\cdots+x^{500}, x>0$, is [JEE Mains-2022]

(A) ${}^{501}C_{101}(5)^{399}$

(B) ${}^{501}C_{101}(5)^{400}$

(C) ${}^{501}C_{100}(5)^{400}$

(D) ${}^{500}C_{101}(5)^{309}$

[Ans. : (A)]

DIBY-38

The coefficient of x^{301} in $(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+\cdots+x^{500}$ is: [JEE Mains-2023]

A.G.P.

(A) ${}^{501}C_{302}$ (B) ${}^{500}C_{301}$ (C) ${}^{500}C_{300}$ (D) ${}^{501}C_{200}$

[Ans. : (D)]

DIBY-39

The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+\cdots+x^{54}(1+x)^{46}$ is ${}^{99}C_p-{}^{46}C_q$. Then a possible value to $p+q$ is: [JEE Mains-2024]

A.G.P.

(A) 55 (B) 61 (C) 68 (D) 83

[Ans. : (D)]

FINDING COEFFICIENTS

DIBY-40

The coefficient of x^7 in the expression $(1+x)^{10}+x(1+x)^9+x^2(1+x)^8+\cdots+x^{10}$ is:

[JEE Mains-2020]

(A) 210 (B) 330 (C) 420 (D) 120

[Ans. : (B)]

DIBY-41

The coefficient of x^{256} in the expansion of $(1-x)^{101}(x^2+x+1)^{100}$ is:

[JEE Mains-2021]

(A) $-{}^{100}C_{15}$ (B) ${}^{100}C_{16}$ (C) $-{}^{100}C_{16}$ (D) ${}^{100}C_{15}$

[Ans. : (D)]

DIBY-42

The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to

[JEE Mains-2024]

[Ans. : (0)]

FINDING COEFFICIENTS

DIBY-43

The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^6$ in powers of x, is

[JEE Mains-2020]

[Ans. : (120)]

DIBY-44

The term independent of x in the expansion of $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}, x \neq 0$ is:

[JEE Mains-2022]

(A) $\frac{7}{40}$ (B) $\frac{33}{200}$ (C) $\frac{39}{200}$ (D) $\frac{11}{50}$

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians